

Annex 4 — Country Studies (PESTEL Analysis)

Purpose: Macro-level analysis of the political, economic, social, technological, environmental, and legal landscape for Urban Rig deployment in each target country. Covers national waste statistics, regulatory frameworks, market conditions, competition, stakeholder mapping, and deployment phasing. Sector-specific feedstock profiles are in Annexes 5–8; financial projections are in Annexes 10–13.

Date: March 2026

4A — France

4A.1 National Waste Statistics

Total MSW (2020): 38.5 million tonnes/year, 580 kg per capita/year — above the EU-27 average of 505 kg (ADEME, *Waste Key Figures (Chiffres-clés Déchets)*, 2022 Edition). Annual growth approximately 0.5–1.0%, declining due to waste prevention policies.

Treatment breakdown (2020):

Method	% of Total	Tonnes/Year (M)	Trend
Recycling (material)	26.9%	10.4	↑ Target: 55% by 2025
Composting / organic recovery	15.8%	6.1	↑ Growing slowly
Incineration with energy recovery	38.1%	14.7	↔ Stable, controversial
Landfilling	18.1%	7.0	↓ Target: <10% by 2035

Source: ADEME 2022; Eurostat.

Key insight: While France reports 80.8% “recovery value” (valorisation: material recovery + composting + energy recovery), only **42.7%** is true material recycling or composting. 38.1% is incineration — which destroys material value and generates CO₂. Approximately **21.6 million tonnes per year** is either burned or buried with no productive recovery pathway under current infrastructure.

4A.2 Île-de-France Regional Data

Population: 12.3 million (2023) — 18% of France. 99% urbanised, generating 31% of French GDP (ORDIF, Regional Waste Observatory for the Île-de-France, 2021).

Waste generation: 6.2 million tonnes/year (2019), 504 kg per capita (lower than national average due to urban density) — approximately 17,000 tonnes/day (ORDIF—Regional Waste Observatory for Île-de-France, 2021).

Regional waste composition:

Component	%	Notes
Organic	28%	Lower than national (less garden waste in apartments)
Paper/cardboard	25%	Higher (office/commercial waste)
Plastics	14%	Higher (packaging, e-commerce)
Glass	11%	—
Metals	3%	—
Textiles	4%	Fashion capital; significant textile waste
Other	15%	Electronics, bulky waste

Source: ORDIF (Regional Waste Observatory for Île-de-France), 2021.

Treatment infrastructure:

- **Incineration:** 14 facilities, ~2.5 M tonnes/year capacity. Notable: Issy-les-Moulineaux (ISSEANE, 460,000 T/yr), Ivry-Paris XIII (730,000 T/yr, being replaced — €1.3B investment), Saint-Ouen (600,000 T/yr). 1,200 GWh electricity + district heating (SYCTOM; ORDIF, 2021).
- **Landfills:** 4 major sites, ~600,000 T/yr capacity (down from 12 sites in 2000). Declining; Île-de-France exports waste to other regions.
- **Recycling/composting:** 10 material sorting centres + 8 composting platforms, ~1.8 M tonnes/year combined.

Treatment breakdown (2019):

Method	Tonnes/Year	%
Incineration	2,300,000	37%
Material recycling	1,550,000	25%
Composting/organics	930,000	15%
Landfilling	1,100,000	18%
Other	320,000	5%

Source: ORDIF (Regional Waste Observatory for Île-de-France), 2021.

4A.3 Regulatory Framework

Anti-Waste Law for a Circular Economy (AGEC Law), 2020

Key targets (Law No. 2020-105 of 10 February 2020):

- Reduce waste generation by **15% by 2030** (vs 2010)
- 100% plastic recycling by 2025 (aspirational)
- End single-use plastics: bags banned (2016), tableware banned (2021), fruit/vegetable packaging <1.5 kg banned (2022)
- Reduce landfilling to **10% by 2035**
- **65% material recycling** by 2035 (EU Directive 2018/851)
- 13 Extended Producer Responsibility (EPR) streams — producers fund collection and recycling

ICPE Classification (Environmental Permitting)

Relevant classifications for Urban Rig (Environmental Code):

- **Category 2770:** Waste Treatment (Classified activity 2770: Waste Treatment). Threshold: >10 T/day = Authorization regime (most stringent)
- **Category 2910:** Combustion (if classified as thermal treatment)

Authorization process (typical 12–24 months):

1. Dossier preparation (6–9 months): environmental impact study (impact assessment), risk assessment (hazard study), technical design files
2. Public inquiry (1–2 months)
3. Regulatory review (6–9 months)
4. Prefectural decree issued if approved

Emission limits (Order of 20 September 2002 [French Order of 20 September 2002], incineration standards — indicative, set case-by-case): particulates <10 mg/Nm³, NO_x <80 mg/Nm³, SO_x <50 mg/Nm³, CO <50 mg/Nm³, dioxins <0.1 ng TEQ/Nm³.

Note: Urban Rig emissions are 5–50× below these limits (see Annex 2 §2.7). The classification as pyrolysis (not incineration) may allow more favourable permitting treatment, but this requires legal confirmation with the Regional Environmental Authority (DREAL—Directions Régionales de l’Environnement, de l’Aménagement et du Logement) Île-de-France.

TGAP (General Tax on Polluting Activities)

Treatment	TGAP (2023)	Trend
Landfilling	€65/T	Rising — strong deterrent
Incineration	€15/T	Rising to €25/T by 2025
Recycling	€0/T	Incentive for circularity

Source: ADEME; Court of Auditors (Cour des Comptes).

Economic incentive to avoid landfilling and incineration drives demand for alternative treatment pathways.

4A.4 Market Conditions

Tipping Fees (Île-de-France, 2023)

Treatment	EUR/T	Notes
Landfilling	140–180	Declining capacity increases prices
Incineration (WTE)	80–120	Most common
Sorting/recycling	60–100	For pre-sorted materials
Composting	40–80	Source-separated organics

Source: ADEME; Court of Auditors (Cour des Comptes).

The financial model uses a blended gate fee of **€114/T** — within the incineration range and well below landfilling costs, making Urban Rig cost-competitive on intake pricing alone.

Product Markets

Product pricing is detailed in Annex 3. Key France-specific market characteristics:

- **Oil buyers:** TotalEnergies (refineries in Île-de-France), Petroineos (Lavéra), industrial users (LafargeHolcim, Vicat). France imports ~60 M tonnes crude oil/year; pyrolysis oil can blend or replace feedstock.
 - **Biochar:** France has 28 M hectares farmland, organic farming at 10% (target: 15% by 2030). Biochar demand currently <5,000 T/year but growing. Industrial fuel market (cement kilns) is more immediately addressable.
 - **Carbon credits:** EU ETS at €50–80/T CO₂ (2024–2025). Waste sector not currently covered but proposed expansion. Voluntary markets: €20–50/T. Financial model uses €50/T conservatively.
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4A.5 Competition

Incumbent Operators

- **SYCTOM** (Paris metropolitan waste agency): Serves 85 municipalities (5.6 M people). Operates 3 incineration plants. Committed to incineration model with large sunk costs — Ivry-Paris XIII replacement alone is €1.3 billion.
- **Suez:** Operates recycling centres and transfer stations. Developing “advanced recycling” (chemical recycling) pilots.
- **Veolia:** Major player in traditional recycling + WTE. Exploring pyrolysis and gasification.

Emerging Technologies

- **Plastic Energy** (UK-based): Pyrolysis plant in Limay (Île-de-France), ~15,000 T/year. Requires clean plastic feedstock only.
- **Carbios** (French biotech): Enzymatic PET recycling. Different technology (biological, not thermal).
- **Enerkem** (Canadian): Gasification of waste to methanol/ethanol. Requires pre-sorted waste.

Urban Rig differentiation: Accepts mixed unsorted waste (vs competitors requiring clean feedstock), continuous operation (vs batch), produces high-value char and oil by-products (see Annex 2 §2.4), modular and faster to deploy than mega-facilities.

4A.6 Stakeholder Mapping

Government: Ministry of Ecological Transition (waste policy), ADEME (grants, technical support), Regional Environmental Authority (DREAL—Directions Régionales de l’Environnement, de l’Aménagement et du Logement) Île-de-France (regional permits), SYCTOM (Paris waste authority), Paris Airports/ADP (CDG operator), municipal governments (waste collection contracts).

Industrial partners: TotalEnergies (oil offtake, potential equity), LafargeHolcim/Vicat (biochar offtake), Suez/Veolia (waste supply, operational partnership).

Financial: Bpifrance (public investment bank, green financing), European Investment Bank (circular economy projects), Crédit Agricole/BNP Paribas (green finance desks).

Research: INERIS (industrial environment and risks), École Polytechnique, UTC Compiègne.

4A.7 Key Success Factors

Favourable:

- Strong policy support (Anti-Waste Law for a Circular Economy, EU directives, TGAP incentives)
- High tipping fees (€80–180/T) make dual revenue model very attractive
- Large addressable market (6.2 M T/year in Île-de-France alone)
- Incineration dependency (37%) creates substitution opportunity
- Declining landfill capacity drives demand for alternatives
- Green financing availability (Bpifrance, EIB, ADEME grants)

Challenges:

- ICPE permitting (12–24 months, comprehensive dossier required)
 - Public acceptance (transparent communication essential)
 - Established incumbents (Suez, Veolia, SYCTOM have infrastructure and long-term contracts)
 - Higher labour costs than global average
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4A.8 Deployment Phasing (France)

Phase 1 — Pilot (Years 1–3): 1 unit at CDG or similar high-profile location. 200 T/day. Prove technology, secure permits, build stakeholder relationships. Partners: Paris Airports (ADP—CDG), ADEME (grant), local waste authority.

Phase 2 — Regional expansion (Years 3–7): 5–10 units across Île-de-France and major cities (Lyon, Marseille, Lille, Toulouse). 1,000–2,000 T/day total. Target: 5% of French incineration capacity.

Phase 3 — National rollout (Years 7–15): 50–100 units nationwide. Become standard solution for residual waste. Integration with existing waste management systems.

4B — South Korea

4B.1 National Waste Statistics

Total MSW (2021): 23.6 million tonnes/year, 457 kg per capita, 64,700 tonnes/day. Annual growth +2–3% (Korea Ministry of Environment — MOE, *2021 Waste Generation and Treatment Status Report*; Korea Environment Corporation — KECO).

Population: 51.7 million (2023). 81.4% urbanisation. 527 people/km² (very high density). Seoul Metropolitan Area: 25.6 million (50% of population).

Treatment breakdown (2021):

Method	% of Total	Tonnes/Year (M)	Trend
Recycling	61.0%	14.4	↑ OECD highest
Incineration	23.6%	5.6	↑ Growing (land scarcity)
Landfilling	11.5%	2.7	↓ Down from 70% in 1995
Composting (organics)	3.1%	0.7	↔ Stable

Source: Korea MOE 2022.

Important caveat on recycling rate: The 61% includes energy-from-waste counted as “energy recycling” and pre-2018 exports to China. True material recycling is closer to **40–45%** (Korea MOE adjusted methodology).

4B.2 Volume-Based Waste Fee (VBWF) System

Korea’s national pay-as-you-throw system (implemented 1995) is a key contextual factor:

- Residents must buy official bags for general waste. Seoul example: 20 L bag = ₩490 (~€0.35), 50 L bag = ₩1,230 (~€0.85) (Seoul Metropolitan Government).
- Separated recyclables are collected free (incentive to sort).
- Food waste: separate paid bags or RFID-weighted bins.
- **Impact:** 30% decrease in general waste after implementation. Recycling rose from 15% (1995) to 61% (2021).

4B.3 Plastic Waste Crisis

Generation: 4.2 million tonnes/year (2021), +50% since 2010, +20% during COVID-19 (delivery food, online shopping) (Korea MOE 2022).

Recycling reality:

Plastic Type	Collection Rate	Actual Recycling Rate	Main Issues
PET bottles	90%	85%	Deposit system success story
PE/PP (clean)	75%	60%	Requires washing
PS (styrofoam)	50%	30%	Contamination, low value
PVC	30%	10%	Toxic concerns
Multi-layer/film	40%	<5%	Not mechanically recyclable
Average	65%	~35% actual	Gap = exported or downcycled

Source: Korea MOE; Korea Recycling Industry Association.

2018 China import ban crisis: Korea exported ~50,000 T/month to China. When China stopped accepting foreign plastic waste in January 2018, waste piled at apartment complexes. Government emergency response included increased domestic recycling capacity, bans on difficult-to-recycle plastics, and exploration of chemical recycling/pyrolysis (Korea MOE).

4B.4 Seoul Metropolitan Area

Seoul city (proper): 9.4 million people, 605 km². Total MSW: 2.6 M T/year (2021), 277 kg per capita (lower than national average due to VBWF success), 7,100 T/day (Seoul Metropolitan Government, 2021 *Seoul Environmental Status Report*).

Seoul waste composition: Recyclables 60%, food waste 20%, general waste 17%, bulky waste 3%.

Treatment infrastructure: 4 major incineration plants with 3,750 T/day combined capacity (Mapo 750, Yangcheon 900, Nowon 600, Gangnam 1,500). Additional capacity at Incheon

and Gyeonggi facilities. Energy recovery: 650 GWh/yr electricity + 2,800 TJ/yr district heating (Seoul Metropolitan Government).

Sudokwon Landfill (Incheon): Largest landfill in the world by daily intake (~15,000 T/day, declining). Serves Seoul + Incheon + Gyeonggi. Opened 1992, planned closure 2044 (4th section). Methane capture generates 50 MW electricity.

4B.5 Regulatory Framework

National Waste Policy

Waste Management Act (enforcement 2016, ongoing amendments): mandatory separation of recyclables, food waste, and general waste. Extended Producer Responsibility (EPR) for packaging, electronics, tires, batteries. No organic waste in landfills since 2005. Strict incineration emission standards (Korea MOE, *5th National Waste Management Plan 2022–2026*).

Targets (2020–2035): Waste reduction 20% per capita by 2030 (vs 2017). Recycling rate: maintain 60%+. Landfilling: <5% by 2030. Single-use plastics: 50% reduction by 2030. Food waste: 30% reduction by 2030.

Environmental Permitting

Pyrolysis/thermal treatment classified as “intermediate waste treatment facility.” >100 T/day requires full licensing + Environmental Impact Assessment (EIA). Licensing authority: provincial government.

Approval process (18–30 months): Pre-consultation with environmental authorities → EIA (6–12 months, including air quality modelling, traffic impact, noise/odour assessment, mandatory public hearing) → Business plan approval by MOE or provincial government → Facility installation permit → Inspection and operation permit after construction (Korea Waste Management Act; MOE guidelines).

Emission Standards

Korean standards are among the world’s strictest, with continuous real-time monitoring (TMS: Tele-Monitoring System):

Pollutant	Limit	Monitoring
Particulates	<10 mg/Nm ³	Continuous
NOx	<70 mg/Nm ³	Continuous
SOx	<30 mg/Nm ³	Continuous
CO	<30 mg/Nm ³	Continuous
Dioxins	<0.1 ng TEQ/Nm ³	Quarterly
HCl	<25 mg/Nm ³	Continuous

Source: Korea Clean Air Conservation Act.

Urban Rig emissions (KANSO Technos report — see Annex 2 §2.7) are well within all Korean limits.

4B.6 Market Conditions

Tipping Fees (Seoul Metropolitan Area, 2023)

Treatment	₩/T	EUR/T	Notes
Landfilling	80,000–120,000	55–83	Limited capacity; increasing
Incineration	60,000–90,000	41–62	Most common for residual waste
Recycling (mixed)	50,000–80,000	34–55	Sorting and processing
Food waste composting	40,000–70,000	28–48	Organics only

Source: Korea Waste Association; exchange rate ₩1,450 = €1 (2024). Note: tipping fees significantly lower than France due to government subsidies and public facility dominance.

Product Markets

- **Oil buyers:** SK Innovation, GS Caltex, S-Oil, Hyundai Oilbank (refineries); LG Chem, Lotte Chemical (petrochemicals, exploring circular feedstocks); Ssangyong, Asia Cement (cement kilns). Korean pricing estimated at ₩700,000–900,000/T (€480–620/T) (industry surveys). Korea imports 100% of crude oil (~140 M T/year) — domestic alternative is strategically valuable.
- **Biochar:** Agricultural market limited (1.5 M hectares farmland, declining, 6% organic). Industrial fuel market more immediately addressable: cement kilns transitioning from coal, ₩100,000–200,000/T (€69–138/T).
- **Carbon credits:** K-ETS launched 2015 (world's 2nd largest national ETS after EU). Carbon price: ₩20,000–40,000/T CO₂ (~€14–28/T) — much lower than EU. Waste sector not currently covered; voluntary markets underdeveloped. **Carbon credits are excluded from Korean financial projections.**

4B.7 Competition

Incumbent Infrastructure

~60 MSW incinerators nationwide, ~18,000 T/day total capacity. Mostly public or public-private partnerships. Municipal operation dominates — private sector has limited penetration (Korea MOE).

Emerging Technologies

- **Dongyang E&P:** Commercial pyrolysis (~10 T/day, plastic waste to oil).
- **SK Geo Centric:** Announced plans for 100,000 T/year chemical recycling facility.
- **Hyosung Chemical:** Depolymerisation of PET (pilot stage).
- **LG Chem:** Researching pyrolysis and solvent-based recycling.
- **International:** Pyrowave (Canada), Agilyx (US) exploring Korean partnerships.

Urban Rig differentiation: Continuous operation (vs batch), accepts mixed unsorted waste (vs clean feedstock), proven technology (vs pilots), produces high-value char and

oil by-products. The high energy content of char makes it valuable on Korean fuel markets. However, a Korean JV partner is essential — foreign companies face barriers in the public-sector-dominated waste market.

4B.8 Stakeholder Mapping

Government: Ministry of Environment (MOE, policy and permits), Korea Environment Corporation (KECO, research and development), Seoul Metropolitan Government (largest waste generator), provincial governments (Gyeonggi, Incheon, Busan — permitting).

Industry: Refineries (SK Innovation, GS Caltex, S-Oil — oil offtake); petrochemicals (LG Chem, Lotte Chemical — circular feedstock); cement (Ssangyong, Asia Cement — biochar offtake); waste management (Doosan Enerbility, SK Ecoplant — potential JV partners); Chaebol (Samsung, Hyundai, LG — ESG initiatives, potential investors).

Financial: Korea Development Bank (KDB, green financing), Korea Investment Corporation (KIC, sovereign wealth fund), commercial banks (KB, Shinhan, Woori — green finance desks), green bond market (growing).

Research: KAIST, Seoul National University, Korea Institute of Industrial Technology (KITECH).

4B.9 Key Success Factors

Favourable:

- Strong waste policy framework (VBWF, separation mandates, landfill phase-out)
- Carbon neutrality goals (waste sector must reduce emissions 50% by 2030)
- High land/facility costs (modular Urban Rig attractive vs mega-incinerators)
- Large addressable market (8–10 M T/year residual waste)
- Sudokwon landfill reaching capacity (drives demand for alternatives)
- Plastic crisis post-China ban (need for domestic solutions)
- NIMBY opposition to incinerators/landfills (creates demand for cleaner technologies)
- Government innovation support (K-Green New Deal, circular economy funding)

Challenges:

- Lower tipping fees (€41–62/T vs €80–120/T in France) — economics rely more heavily on product sales
 - Lengthy permitting (18–30 months with mandatory EIA and public hearings)
 - Low carbon price (~€14–28/T vs €50–80/T EU ETS)
 - Public sector incumbency (municipalities operate WTE; may resist private alternatives)
 - Emerging local competitors developing pyrolysis technology
 - Preference for Korean technology — JV with Korean partner essential
 - Public scepticism of “waste-to-energy” label (past incinerator incidents; transparency critical)
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4B.10 Deployment Phasing (Korea)

Phase 1 — Pilot/demonstration (Years 1–3): 50–100 T/day pilot in an industrial waste park (Sihwa, Ulsan) or R&D zone. Partners: Korean waste/environmental JV partner (Doosan Enerbility, SK Ecoplant, or KECO subsidiary) + MOE (K-Green New Deal funding). Goals: demonstrate emissions/yields/uptime, secure regulatory approval, establish product markets (oil → refinery,

biochar → pilot users). Estimated cost: ~€37M (URC-1000 unit ~€30M + site + accessories + contingency). Funding: 50% equity (JV), 30% government grant/loan, 20% project finance.

Phase 2 — Commercial deployment (Years 3-7): 1-2 units × 200 T/day in Seoul Metropolitan Area (Incheon, Gyeonggi). PPP model with city government. Long-term waste supply contract (15 years). Oil offtake pre-negotiated with SK Innovation or GS Caltex. Estimated cost: ~€124M (2 × URC-2000). Funding: PPP structure (government equity + private finance + green bonds).

Phase 3 — National expansion (Years 7-15): 10-20 units across major cities. Target: 1.5-2 M T/year (20-25% of addressable market). Potential to export model to Japan, Taiwan, and other Asian markets.

Total Korea potential: 15-30 units over 10-15 years.

4C — Country Comparison Summary

Factor	France	South Korea
MSW volume	38.5 M T/year	23.6 M T/year
Recycling rate (true material)	~43%	~40-45%
Incineration dependency	38.1%	23.6%
Tipping fees	€80-180/T	€41-83/T
Gate fee assumption (model)	€114/T	€55/T
Carbon credit price	€50-80/T (EU ETS)	€14-28/T (K-ETS)
Permitting timeline	12-24 months	18-30 months
Labour cost (15 FTE, loaded)	~€1.2M/year (~€83K/person)	~€0.83M/year (~€55K/person)
Oil price scenario	Medium: €641/T	~€550/T
CAPEX (URC-2000)	€60.8M	€61.8M (higher land)
Revenue (steady state)	€17.3M/year	€11.8M/year
EBITDA (margin)	€11.9M (69%)	€5.7M (48%)
Payback	~5.4 years	~10.8 years
Key driver	Gate fees (31%) + oil (41%)	Oil (65%)

Sources: Financial projections from UR-Financial-Model-General.xlsx (France) and Korea financial model worksheet (Korea). Detailed scenario economics in Annexes 10-13.

France offers superior near-term economics driven by high tipping fees, carbon credit eligibility, and the regulatory push away from incineration. Korea offers a larger long-term addressable market and strategic proximity to the UR manufacturing base, but economics depend more heavily on oil sales and would benefit significantly from K-ETS expansion to the waste sector.

4D — References

France — Institutional and Regulatory Sources

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2. **Eurostat** — EU waste management statistics. Cited for: EU-27 per capita averages (§4A.1).
3. **ORDIF (Regional Waste Observatory for Île-de-France)** — 2021 report. Cited for: Île-de-France population, waste generation, composition, and treatment infrastructure (§4A.2).
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8. **SYCTOM** — Paris metropolitan waste agency. Cited for: incineration infrastructure data, Ivry-Paris XIII project budget €1.3B (§4A.2, §4A.5).
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Korea — Institutional and Regulatory Sources

10. **Korea Ministry of Environment (MOE)** — *2021 Waste Generation and Treatment Status Report (2021 Waste Generation and Treatment Status); 5th National Waste Management Plan 2022–2026*. Cited for: national waste statistics, treatment breakdown, recycling rates, plastic waste crisis, and permitting requirements (§4B.1, §4B.3, §4B.5, §4B.7).
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16. **Korea Waste Association** — Tipping fee data. Cited for: treatment cost benchmarks (§4B.6).
17. **K-ETS (Korea Emissions Trading Scheme)** — Carbon pricing (launched 2015). Cited for: carbon credit pricing ₩20,000–40,000/T CO₂ (§4B.6).

Internal Model References

18. **UR-Financial-Model-General.xlsx** — France financial projections. Cited for: country comparison table (§4C).
19. **Korea financial model worksheet** — Korea financial projections. Cited for: country comparison table (§4C).
20. **Hardin LOI** — Letter of Intent for 2 × URC-1000 at USD 60M. Cited for: Korea Phase 1 pricing (§4B.10, via Technical Findings Log §16.1).

End of Annex 4